
IAL Chemistry Unit 4

Rapid Revision

WCH14 — Rates, Equilibria and Further Organic Chemistry

Last-minute cram sheets — designed to be scanned, not read

Condensed from the full A* guide (21 past papers, Jun 2019 - Jun 2026)

90 marks · 1 h 45 min · $\approx$ 1.1 min per mark

Section A: 20 MCQ (**$\leq$ 20 minutes**) · Section B: ~48–52 · Section C: ~18–22

A* = grade A overall ($\geq$ 480/600) AND $\geq$ 270/300 across Units 4, 5 and 6.

TRIAGE — ON VIRTUALLY EVERY PAPER

An **entropy / feasibility calculation** · a **Born-Haber or solution cycle** · an **initial-rates table** → **rate equation** → **k with units** · an **Arrhenius or half-life graph** · **K_p or K_c from moles** · **weak-acid pH and buffer pH** · the **HCN + carbonyl mechanism with the racemic explanation** · **carbonyl tests** · **NMR deduction**.

Priority order: (1) entropy chain and feasibility · (2) K_p/K_c · (3) weak acid, buffer and excess-base pH · (4) orders, k units, Arrhenius · (5) HCN mechanism + racemic · (6) Born-Haber, Hess, polarisation · (7) carbonyl tests, acyl chlorides, esters · (8) NMR deduction.

THE 6-MARK LEVELS QUESTION

6 indicative points = 4 marks + 2 for reasoning. 5–4 points = 3 + 1 if partly linked. 3–2 points = 2 + 0 if unstructured. **Incorrect chemistry loses the reasoning marks.**

Write **six separate points, each a full sentence, in a logical order.**

THE ROUTINE FOR EVERY QUESTION

Calculations: expression → substitution → answer → **UNITS** → s.f. Check: **J vs kJ** · **K vs °C** · **×2 for diprotic or two ions** · **divide by V**.

Explanations: one idea per sentence, with the key word — *reservoir* · *planar C=O* · $E \geq E_a$ · *charge density* · *K_p unchanged*.

Mechanisms: dipoles · lone pairs · arrows **from lone pairs or bonds** · charges on ions and intermediates.

SHEET 1 — ORDERS, RATE EQUATIONS & GRAPHS

ORDERS FROM AN INITIAL-RATES TABLE

1. Pick two experiments where **only one** concentration changes. $\times 2 \rightarrow$ rate $\times 1$ (order 0), $\times 2$ (order 1), $\times 4$ (order 2).
2. When two change at once, **divide out** the effect of the one you know.
3. **Quote the experiment numbers and the factors** — "orders without justification" scores 1.
4. Write rate = $k[A]^m[B]^n$ — **square brackets** (round brackets are penalised); include k.

Overall order	Units of k
0	$\text{mol dm}^{-3} \text{ s}^{-1}$
1	s^{-1}
2	$\text{dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$
3	$\text{dm}^6 \text{ mol}^{-2} \text{ s}^{-1}$
4	$\text{dm}^9 \text{ mol}^{-3} \text{ s}^{-1}$

Units usually earn their own mark. 1 s.f. answers are penalised.

Recurring: rate = $k[\text{BrO}_3^-][\text{Br}^-][\text{H}^+]^2$ ($\text{dm}^9 \text{ mol}^{-3} \text{ s}^{-1}$) · $k[\text{NO}]^2[\text{O}_2]$ ($\text{dm}^6 \text{ mol}^{-2} \text{ s}^{-1}$) · $k[\text{H}_2\text{O}_2][\text{I}^-]$ · propanone iodination $k[\text{CH}_3\text{COCH}_3][\text{H}^+]$, **zero order in I_2** .

Graph	Zero order	First order	Second order
conc vs time	straight line down	constant half-life	half-life increases
rate vs conc	horizontal line	straight line through the origin	upward curve

DON'T SWAP THE TWO PIECES OF EVIDENCE

"**Constant half-life**" is evidence from a **concentration-time** graph. "**Straight line through the origin**" is evidence from a **rate-concentration** graph.

Initial rate: draw a **tangent at $t = 0$** and find its gradient — "gradient of the graph" alone is rejected. **Half-life:** show **two successive half-lives on the graph** (working on the graph is required); equal $\rightarrow$ first order. **$k = \ln 2 \div t_{1/2} = 0.693 \div t_{1/2}$ — convert minutes to seconds.**

Excess reagent: "its concentration **stays effectively constant**, so the rate depends only on [the other reactant]".

Following a reaction: colorimetry (a coloured species) · gas volume / mass loss · pressure change (**not** for $\text{H}_2 + \text{I}_2 \rightarrow 2\text{HI}$, no change in moles) · conductivity · **quench and titrate** (quench with **NaHCO_3** to remove the H^+ catalyst — **not NaOH** , which also catalyses and reacts with I_2) · clock reaction, rate $\propto 1/t$. **Iodine clock:** a **small fixed** amount of thiosulfate reacts with I_2 as it forms; when used up, I_2 + starch $\rightarrow$ **blue-black** (not "clear"). Kept small so **other concentrations barely change**, so $1/t$ measures the **initial** rate.

SHEET 2 — MECHANISMS & ARRHENIUS

The rate equation contains the species in, or before, the rate-determining step, with the numbers of each. A species **not** in it (I_2 in propanone iodination) reacts **after** the slow step. A one-step reaction is unlikely with 3+ particles: "**a collision of three or more particles at once is very unlikely**" (reject "**three moles/compounds**"). A proposed mechanism must **add up to the overall equation** and have an **RDS matching the rate equation**. **Heterogeneous catalysis:** reactants **ADSORB** (not "absorb") $\rightarrow$ bonds weakened $\rightarrow$ products **desorb**.

	$\text{S}_{\text{N}}2$ — primary	$\text{S}_{\text{N}}1$ — tertiary
Rate equation	rate = $k[\text{RX}][\text{OH}^-]$	rate = $k[\text{RX}]$
Steps	One step via a transition state $[\text{HO}\cdots\text{C}\cdots\text{X}]^-$	Slow $\text{RX} \rightarrow \text{R}^+ + \text{X}^-$, then fast $\text{R}^+ + \text{OH}^-$
Why	Little steric hindrance	Stable tertiary carbocation
Chiral substrate	Inversion — attack from the side opposite the leaving group $\rightarrow$ product optically active	Planar carbocation , attacked equally from either side $\rightarrow$ racemic mixture
Energy profile	One hump	Two humps; trough = intermediate; first E_{a} larger = RDS

A mechanism must match your own rate equation — a mismatch caps the marks.

WHY RATE RISES WITH T — 3 MARKS, AND WHAT IS IGNORED

k increases · average kinetic energy increases · a greater proportion of collisions have **$E \geq E_{\text{a}}$** , E_{a} itself is **unchanged**.

"**IGNORE more successful collisions / more energy / more frequent collisions.**"

ARRHENIUS — $\ln k = -E_{\text{a}}/(RT) + \ln A$

Plot **$\ln k$ (y, no units) against $1/T$ (x, K^{-1})**; straight best-fit line; $\geq$ **half the grid**; points $\pm \frac{1}{2}$ square.

Gradient = $-E_{\text{a}}/R$ (units K, negative) $\rightarrow E_{\text{a}} = -\text{gradient} \times R \rightarrow$ give in **kJ mol^{-1} with a + sign**.

Two-temperature form: **$\ln(k_2/k_1) = (E_{\text{a}}/R)(1/T_1 - 1/T_2)$** .

A **negative E_{a}** or a **positive gradient** loses the mark. **Kinetic stability:** feasible ($\Delta S_{\text{total}} > 0$) but **high activation energy** $\rightarrow$ too slow at room temperature.

SHEET 3 — BORN-HABER, LATTICE ENERGY & SOLUTION

Lattice energy: when **1 mol of ionic solid** forms **from its gaseous ions** (exothermic). **Atomisation:** forms **1 mol of gaseous atoms** — $\frac{1}{2}\text{Cl}_2(\text{g}) \rightarrow \text{Cl}(\text{g})$. **Electron affinity:** 1 mol of gaseous atoms each gain an electron; the **second EA is endothermic** — an electron is added to a **negative ion, so there is repulsion**. **Hydration enthalpy:** when 1 mol of **gaseous ions** is completely hydrated — **always exothermic (reject "energy required")**.

DRAWING THE CYCLE — AND THE CLASSIC ERROR

Upward arrows: atomisation, IE_1 , IE_2 . **Downward:** EA_1 , LE, $\Delta_f H$.

Every level needs **species with state symbols AND electrons**: $\text{Ca}^{2+}(\text{g}) + 2\text{Cl}(\text{g}) + 2\text{e}^-$. **Penalised once:** missing state symbols, missing electrons, writing the change on the line.

$\Delta_f H = \Delta_{\text{at}} H(\text{M}) + \text{IEs} + n \times \Delta_{\text{at}} H(\text{X}) + n \times \text{EA} + \text{LE}$

The classic error is forgetting $\times 2$ for 2Cl atoms / 2 EAs — on some schemes it is the only TE allowed.

Why MgCl_2 doesn't form: IE_3 of Mg is huge (an electron from an inner shell), not repaid by the extra lattice energy.

LATTICE ENERGY AND POLARISATION — 3 MARKS

LE is **more exothermic when ions are smaller (ionic radius, not atomic) and more highly charged** → stronger electrostatic attraction.

Theoretical vs experimental: the theoretical value assumes **100% ionic, perfect spheres / point charges**. If the **Born-Haber value is more exothermic than theoretical → covalent character**. Cause: a **small, highly charged cation (high charge density)**

polarises/distorts the electron cloud of a large, easily polarised anion.

Close values → almost 100% ionic (NaCl, NaF, MgO — O^{2-} small, hard to polarise). **Big difference → covalent character** (MgI_2 , MgCl_2 , AgBr, MgS).

Wordings traps: write **ions not atoms** ("iodide", not "iodine") · never " **CaI_2 is covalent**" — it has **covalent character** · never "intermolecular forces" · **use comparatives**.

$\Delta_{\text{sol}} H = \Sigma \Delta_{\text{hyd}} H - \text{LE}$. Multiply by the number of each ion (2Cl^-); to get **one** $\Delta_{\text{hyd}} H$, divide by the number of ions. Hydration is **more exothermic for ions of higher charge / smaller radius** (charge density) → stronger **ion-dipole** attraction to water (**not "intermolecular forces"**). Calorimetry: $q = mc\Delta T$ with the **mass of water only**.

Endothermic yet it dissolves (KCl, NH_4NO_3): **ΔS_{system} is positive enough to make ΔS_{total} positive.**

SHEET 4 — ENTROPY & FEASIBILITY (the top priority)

Entropy = a measure of **disorder / the number of ways of distributing energy among particles**. S increases **solid < liquid < gas**, with more atoms/complexity, and with temperature (jumps at melting, bigger at boiling). **$S^\circ = 0$ only for a perfect crystal at 0 K.**

Predicting the sign of ΔS_{system} (2 marks): compare **moles of gas (or states) WITH NUMBERS** — "1 mol solid → 3 mol gas", "gas moles fall 4 → 2". **A statement with no comparison is rejected.**

Step	Formula	Traps
ΔS_{system}	$\Sigma S(\text{products}) - \Sigma S(\text{reactants})$	Multiply by coefficients; $\text{J K}^{-1} \text{mol}^{-1}$
$\Delta S_{\text{surroundings}}$	$-\Delta H \div T$	ΔH in J ($\times 1000$); T in K (+273)
ΔS_{total}	$\Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$	Mixing J and kJ is rejected
Feasible?	$\Delta S_{\text{total}} > 0$	
Minimum T	Set $\Delta S_{\text{total}} = 0 \rightarrow T = \Delta H \div \Delta S_{\text{system}}$	ROUND UP (661.5 K → 662; 661 rejected)

Gibbs: $\Delta G = \Delta H - T\Delta S_{\text{system}}$; **$\Delta G = -T\Delta S_{\text{total}}$** ; feasible if $\Delta G < 0$. Endothermic $\Delta H \rightarrow \Delta S_{\text{surr}}$ **negative**; exothermic → positive.

ENTROPY AND K

$\Delta S_{\text{total}} = R \ln K$ · $\Delta G = -RT \ln K$ · $K = e^{(\Delta S_{\text{total}} / R)}$

ΔS_{total} positive → **$K > 1$** ; negative → **$K < 1$** . **No TE on a wrong expression.**

ΔS_{total} vs $1/T$ graph: gradient = **$-\Delta H$** ; where it crosses $\Delta S_{\text{total}} = 0$ is the temperature at which **feasibility changes**.

Effect of raising T (3 marks — Le Chatelier alone is IGNORED): 1. ΔS_{system} and ΔH are roughly **constant** with T · 2. **$\Delta S_{\text{surr}} = -\Delta H/T$ becomes less positive** (exothermic) · 3. so **ΔS_{total} decreases** → since $\Delta S_{\text{total}} = R \ln K$, **K decreases**.

ΔH	ΔS_{system}	Feasible?
− (exo)	+	Always
−	−	Only at low T (when $-\Delta H/T > \Delta S_{\text{sys}} $)
+ (endo)	+	Only at high T (when $\Delta S_{\text{sys}} > \Delta H/T$)
+	−	Never

"Always feasible" (3 marks): ΔH negative → ΔS_{surr} always positive; ΔS_{system} positive → **ΔS_{total} always positive.**

SHEET 5 — K_c AND K_p

THE BRACKET RULE — AN AUTOMATIC MARK EITHER WAY

K_c uses square brackets. K_p uses p(...) — round or no brackets; **square brackets are rejected for K_p. Powers go outside the brackets.**

Omit solids (and pure liquids): $\text{C(s)} + \text{H}_2\text{O} \rightleftharpoons \text{CO} + \text{H}_2 \rightarrow \text{K}_c = [\text{CO}][\text{H}_2]/[\text{H}_2\text{O}]$.

Fractional equations give fractional powers. Halving the equation → **K becomes $\sqrt{\text{K}}$** ; doubling → **K²**.

Units: work them out from the expression — atm⁻², atm⁻¹, atm, Pa⁻¹, mol dm⁻³... **No units when they cancel** (esterification K_c; $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$).

Dynamic equilibrium: concentrations **stay constant**; forward and backward **rates are equal** (reject "concentrations equal").

K_c from moles: initial moles → change (equation ratio) → equilibrium moles → **divide by the volume** (forgetting ÷V is the usual loss) unless $\Delta n = 0$ → substitute → units.

K_p from moles (4-5 marks): equilibrium moles → **total moles** → **mole fraction $x = n \div n(\text{total})$** → **partial pressure $p = x \times P(\text{total})$** → substitute → K_p with units. **Omitting the mole-fraction or partial-pressure step typically costs 1 mark.** Don't round early.

K_p = K_c(RT)^{Δn}. Q_p vs K_p: if the quotient ≠ K_p it is **not at equilibrium**.

Change	Effect on K	Effect on position / yield
Temperature	K changes — exothermic forward: T ↑ → K decreases	Follows K
Pressure	None	Shifts to fewer gas moles — state the numbers (3 → 2)
Concentration	None	Shifts to restore K
Catalyst / surface area	None	None — forward and backward rates increase equally

PRESSURE ARGUED THROUGH K_p — "LE CHATELIER" ALONE IS IGNORED

1. K_p is unchanged (only T changes K) · **2. increasing P increases the denominator more than the numerator** (more gas moles on the left), so the **quotient falls below K_p** · **3. the equilibrium shifts right until the quotient again equals K_p** → more product.

Temperature and K: say "**K decreases because the forward reaction is exothermic**" — "shifts left" alone **doesn't earn the K mark**.

Large K → "the equilibrium **lies** far to the right" (not "shifts right").

SHEET 6 — ACIDS, BASES & BUFFERS

Acid = proton donor, base = proton acceptor. Strong = fully dissociated; weak = partially dissociated — not about concentration. Electron-withdrawing groups (Cl, OH) stabilise the anion / weaken O–H → **stronger acid** ($\text{CHCl}_2\text{COOH} > \text{CH}_2\text{ClCOOH} > \text{CH}_3\text{COOH}$). **Diprotic:** H_2SO_4 's first dissociation is complete, and the H^+ produced suppresses the second.

Type	Method	Watch for
Strong acid	$\text{pH} = -\log[\text{H}^+]$	Negative pH is possible; diprotic $\times 2$
Strong base	$[\text{H}^+] = K_w/[\text{OH}^-]$	Ba(OH)₂, Ca(OH)₂, Sr(OH)₂ → [OH[−]] = 2c ; use the given K_w at other temperatures
Dilution (strong)	$\times 10$ volume → pH changes by 1	Water to add = new volume – old
Weak acid	$[\text{H}^+] = \sqrt{K_a \times [\text{HA}]}$	Convert $\text{g dm}^{-3} \rightarrow \text{mol dm}^{-3}$ first; pH must be between 1 and 7
Excess strong base/acid	moles excess ÷ total volume	2 × moles H_2SO_4

WRITE K_a IN FULL

$K_a = [\text{H}^+][\text{A}^-]/[\text{HA}]$, with the **actual formulae**, square brackets, **no [H₂O]**. Writing only $[\text{H}^+]/[\text{HA}]$ as the expression has been rejected.

The two weak-acid assumptions: **1.** dissociation is negligible, so $[\text{HA}]_{\text{eq}} = [\text{HA}]_{\text{initial}}$ · **2.** $[\text{H}^+]$ from water is negligible, so $[\text{H}^+] = [\text{A}^-]$.

When the acid is **fairly strong**, assumption 1 fails: $[\text{HA}]_{\text{eq}}$ is **lower** than assumed → calculated $[\text{H}^+]$ **too high** → calculated pH **too low**.

On $\times 10$ dilution: HCl pH rises by 1; CH_3COOH by only ~ 0.5 , **because the weak acid dissociates more on dilution**.

K_w and temperature: ionisation of water is **endothermic** → higher T, **K_w increases, pH < 7 — but water is still neutral** ($[\text{H}^+] = [\text{OH}^-]$).

BUFFERS — "RESISTS", NEVER "PREVENTS"

A solution that **resists changes in pH** when **small amounts** of acid or alkali are added.

How it works (3–4 marks): **1.** a **large reservoir** of HA and A^- · **2.** added H^+ : $\text{A}^- + \text{H}^+ \rightarrow \text{HA}$ · **3.** added OH^- : $\text{HA} + \text{OH}^- \rightarrow \text{A}^- + \text{H}_2\text{O}$ · **4.** the ratio $[\text{HA}]/[\text{A}^-]$ **hardly changes**, so the pH hardly changes.

Use the actual species — a generic HA/A[−] answer is capped. An ionic equation is required.

pH = $\text{p}K_a + \log([\text{A}^-]/[\text{HA}])$, or $[\text{H}^+] = K_a \times [\text{HA}]/[\text{A}^-]$. **Partial neutralisation:** acid left = acid – OH^- added; salt formed = OH^- added (forgetting the subtraction costs 1). **Half-neutralisation: pH = $\text{p}K_a$. Inverting the ratio is the usual error.**

Titration	Equivalence pH	Indicator
Strong acid – strong base	7	Any (methyl orange or phenolphthalein)
Weak acid – strong base	>7 (the anion is hydrolysed : $\text{A}^- + \text{H}_2\text{O} \rightleftharpoons \text{HA} + \text{OH}^-$)	Phenolphthalein
Strong acid – weak base	<7 ($\text{NH}_4^+ + \text{H}_2\text{O} \rightleftharpoons \text{NH}_3 + \text{H}_3\text{O}^+$)	Methyl orange / methyl red
Weak acid – weak base	~ 7 , no vertical section	None suitable — no sharp change

An indicator's **pH range must lie entirely within the vertical section**. Indicators are weak acids: $\text{HIn} \rightleftharpoons \text{H}^+ + \text{In}^-$, the two forms having different colours.

Diprotic acid: two vertical sections (V and 2V); **$\text{p}K_{a1}$ and $\text{p}K_{a2}$ at the two half-equivalence points.**

SHEET 7 — STEREOCHEMISTRY & THE HCN MECHANISM

Chiral centre: a carbon with **four different atoms or groups**; the molecule has a **non-superimposable mirror image** (reject "four different molecules").
Optically active: rotates the plane of plane-polarised light (reject "bends / deflects"); enantiomers rotate it in **opposite directions**. **Racemic mixture:** equal amounts (50:50) of the two enantiomers → rotations cancel, **no net rotation** (not just "two enantiomers"). **n chiral centres → up to 2ⁿ stereoisomers**.

NUCLEOPHILIC ADDITION OF HCN — THE MOST TESTED MECHANISM

Conditions: HCN with KCN (pH ~5–8). **Why KCN:** HCN is a **weak acid**, so on its own [CN⁻] is **very low**; CN⁻ is the **nucleophile in the slow step**. KCN is a **homogeneous catalyst** — same phase, CN⁻ regenerated.

The seven mark points: 1. lone pair on the **C** of :CN⁻ with its negative charge — **not from N** · 2. curly arrow from that lone pair to the **carbonyl C^{δ+}** · 3. **δ⁺/δ⁻ on C=O** · 4. arrow from the **C=O π bond to O** · 5. **alkoxide intermediate with a FULL negative charge on O** (not δ⁻) and a lone pair · 6. arrow from the O⁻ lone pair to **H of HCN** · 7. arrow from the **H-CN bond to C of CN**; product.

Arrows must start at a lone pair or bond, not at a charge. Half-arrowheads penalised once; **extra wrong arrows cancel right ones**.

WHY THE PRODUCT IS RACEMIC — THE EXACT REJECTIONS

The **C=O group is trigonal planar** — say "**planar carbonyl group**" (reject "**planar molecule / intermediate / carbocation**") · **CN⁻ attacks equally from above and below** (reject "**from any side**") · so **equal amounts of the two enantiomers**, hence no optical activity. **NEVER mention S_N1 / S_N2 here — it loses marks.**

Symmetrical ketones (propanone, pentan-3-one) give an **achiral product** — "racemic" is **wrong** for them. **Enzymes are stereospecific** → a single enantiomer.

Follow-on: hydrolyse with **dilute HCl, reflux**: CH₃CH(OH)CN + 2H₂O + H⁺ → CH₃CH(OH)COOH + NH₄⁺ (**NH₃ instead loses a mark**).

Test	Positive result	Shows
2,4-DNPH (Brady's)	yellow/orange/red ppt (reject brown)	C=O — aldehyde or ketone
Tollens'	silver mirror	aldehyde
Fehling's/Benedict's	blue → brick-red ppt	aldehyde
Acidified K ₂ Cr ₂ O ₇ , warm	orange → green	aldehyde, or 1°/2° alcohol
I₂ + NaOH (both needed)	pale yellow ppt CHI ₃ , antiseptic smell	CH₃CO- or CH₃CH(OH)-
NaHCO ₃ / Na ₂ CO ₃	effervescence	-COOH
PCl ₅	misty/steamy fumes	-OH (alcohol or acid)

Identifying a carbonyl by its derivative: react with 2,4-DNPH → filter (Büchner) → **recrystallise** (minimum hot solvent, cool, filter, **rinse with ice-cold solvent**, dry) → **measure the melting temperature and compare with data**. **Never boiling temperature**; the m.p. is **of the derivative**, not the carbonyl.

Reduction: LiAlH₄ in dry ether (reject aqueous) or NaBH₄. Aldehyde → 1° alcohol; ketone → 2° alcohol; acid → 1° alcohol (LiAlH₄ only). **C=C is not reduced. A tertiary alcohol can never be made by reduction.**

Why C=O is attacked by nucleophiles but C=C isn't: C=O is **polar**, so nucleophiles are attracted to the δ⁺ carbon; **C=C is non-polar and its π electrons repel nucleophiles**.

SHEET 8 — ACIDS, ACYL CHLORIDES, ESTERS & POLYMERS

Carboxylic acids: hydrogen bond with **each other and with water**; long chains lower solubility. With carbonates: $\text{RCOOH} + \text{NaHCO}_3 \rightarrow \text{RCOONa} + \text{H}_2\text{O} + \text{CO}_2$ — the salt is **ionic (reject a covalent O-Na bond)**. With $\text{NH}_3(\text{aq})$: the **ammonium salt, not an amide**. With PCl_5 → acyl chloride + POCl_3 + HCl (misty fumes).

Acyl chlorides react **vigorously at room temperature, no catalyst**, releasing HCl (misty fumes; white smoke with NH_3). Mechanism: **addition-elimination**. + water → carboxylic acid · + alcohol → **ester** · + ammonia → **amide** · + primary amine → **N-substituted amide** · tertiary amines **do not react**.

ACYL CHLORIDE vs ACID + ALCOHOL FOR ESTERS — EACH POINT NEEDS ITS EXPLANATION

Advantages: fast, at room temperature → **lower energy cost** · **not an equilibrium, goes to completion** → higher yield · no catalyst → easier purification.

Disadvantages: HCl is toxic/corrosive → fume cupboard, corrosion-resistant plant · the acyl chloride **must be made first, anhydrous** → extra step, **lower atom economy**.

Esters: named **alkyl group from the alcohol first**, then the acid part ($\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3$ = **propyl ethanoate**). Formation: acid + alcohol, **conc. H_2SO_4 catalyst** (the role is **catalyst**, not "proton donor"), heat — **reversible**. **Alkaline hydrolysis:** NaOH , reflux → **carboxylate salt + alcohol (not the acid)**. **IR: ester $\text{C}=\text{O}$ 1750–1735, no broad $\text{O}-\text{H}$** . Esters **don't react with NaHCO_3** .

Condensation polymers: polyester = diol + dicarboxylic acid (or diacyl chloride), losing H_2O or HCl · **polyamide** = diamine + diacyl chloride/acid. **Repeat unit:** show one unit with **continuation bonds out of both ends; check the O count** — a common MCQ distractor adds an extra O. **M_r of a polymer from n monomers = $n \times (\text{monomer} - 18) + 18$** .

SHEET 9 — NMR, IR, MS & CHROMATOGRAPHY

^{13}C NMR: number of peaks = number of carbon environments — use **symmetry**. Shifts: $\text{C}=\text{O}$ **160–225** (ketone/aldehyde 190–225; acid/ester ~170) · $\text{C}-\text{O}$ 50–90 · $\text{C}-\text{C}$ 0–60. propanone 2 peaks · pentan-3-one 3 · propanal 3 · 3-bromopentane 3.

^1H NMR pattern	Means
Triplet (3H) + quartet (2H)	CH_3CH_2-
Doublet (6H) + septet (1H)	$(\text{CH}_3)_2\text{CH}-$
9H singlet	$(\text{CH}_3)_3\text{C}-$
Two triplets	$-\text{CH}_2\text{CH}_2-$ between groups
Singlet	No H on the adjacent carbon — $\text{CH}_3\text{CO}-$, $\text{CH}_3\text{O}-$, $\text{O}-\text{H}$

Splitting = $n + 1$, where n = H on **adjacent** carbons. **Area (integration) = number of H** in each environment.

Shifts: $\text{R}-\text{CH}_3$ / $\text{R}-\text{CH}_2-$ 0.8–1.8 · $\text{H}-\text{C}-\text{C}=\text{O}$ 1.6–2.9 · **$\text{H}-\text{C}-\text{O}$ 2.8–4.2** · $\text{H}-\text{C}-\text{Br}$ 3.7–4.2 · $\text{O}-\text{H}$ variable ~1–5 (singlet) · **$-\text{CHO}$ 9–10** · **$-\text{COOH}$ 10–12**.

DEDUCTION WRITE-UPS (5–6 MARKS) — DATA → INFERENCE, EVERY LINE

"4 peaks → **4 H environments**" · "areas **3:2:2:3**" · "quartet at 4.1 → **CH_2 next to CH_3 and bonded to O**" · "singlet at 2.2 → **CH_3CO with no adjacent H**" → then the structure.

Say **"the H on C..."**, not the functional group name.

Distinguishing isomers (a 6-mark question): give the **number of peaks, areas, splitting and a shift for each compound**, plus **one MS difference** (propanal m/z 29; propanone 43 only).

MS fragments: 15 CH_3^+ · **29 C_2H_5^+ or CHO^+** · 43 CH_3CO^+ or C_3H_7^+ · 45 COOH^+ · 57 $\text{C}_2\text{H}_5\text{CO}^+$ · **$M - 15$ = loss of CH_3** . High-resolution MS distinguishes formulae with the same integer mass (C_3H_8 44.0632 vs $\text{C}_2\text{H}_4\text{O}$ 44.0265 vs CO_2 43.9898).

IR: $\text{O}-\text{H}$ alcohol 3750–3200 (broad) · **$\text{O}-\text{H}$ acid 3300–2500 (very broad)** · **$\text{C}=\text{O}$ ester 1750–1735** · aldehyde ~1740–1720 · ketone 1720–1700 · acid 1725–1700 · **acid anhydride gives TWO peaks (~1820 and ~1750)** · $\text{C}=\text{C}$ 1669–1645. **Ignore the fingerprint region.**

Chromatography: **R_f = distance moved by the spot ÷ distance moved by the solvent front**. Higher R_f → weaker attraction to the stationary phase / more soluble in the mobile phase. GC: inert carrier gas, **peak area ∝ amount**. HPLC with a **non-polar stationary phase: the most polar compound elutes first**.

SHEET 10 — THE 20 MOST LOST MARKS

1. Forgetting **x2** (two Cl atoms / two EAs / hydration of 2Cl^-) in Born-Haber and Hess cycles.
2. **kJ not converted to J** in $\Delta S_{\text{surr}} = -\Delta H/T$; **°C not converted to K**.
3. **Stoichiometric coefficients missed** in ΔS_{system} .
4. **Rounding the minimum temperature down** instead of up.
5. Not **dividing by volume** for Kc; skipping **mole fractions or partial pressures** for Kp.
6. **Square brackets in Kp**; round brackets in Kc / Ka / rate equations.
7. **Missing or wrong units of k**; k in min^{-1} instead of s^{-1} .
8. **Negative E_a** , or E_a left in J when kJ is asked.
9. **Orders stated without quoting the experiments**.
10. "More successful collisions" instead of "**more collisions with $E \geq E_a$** ".
11. Forgetting **$[\text{OH}^-] = 2c$** for $\text{Ba}(\text{OH})_2/\text{Ca}(\text{OH})_2/\text{Sr}(\text{OH})_2$; forgetting **$2 \times \text{H}_2\text{SO}_4$** .
12. Excess-base concentration **not over the total volume**.
13. **Inverting the buffer ratio**.
14. "Buffer keeps pH **constant**" instead of "**resists change**"; **no ionic equations**.
15. Temperature effect on K explained only by "shifts left" **with no mention of K**.
16. Pressure answered by Le Chatelier when the question asks **through Kp**.
17. "Planar **intermediate / molecule**" instead of "**planar $\text{C}=\text{O}$** "; **mentioning $\text{S}_{\text{N}}1/\text{S}_{\text{N}}2$ in HCN addition**.
18. Curly arrow **from N of CN^-** , or **from a charge** instead of a lone pair; **no charge on the alkoxide**.
19. Polarisation answers using **atoms**, "**covalent**" instead of "**covalent character**", or **no comparison**.
20. " LiAlH_4 " **without dry ether**; "distil" **without "as it forms"**; "**boiling** temperature" of the derivative.

EXAMINER RULINGS WORTH ONE LAST READ

Sig figs: "ignore s.f. **except 1 s.f.**" is the default — **1 s.f. always loses a mark**. Obey explicit demands. **Round minimum temperatures up. Don't round early.**

Units on k, Kc/Kp, ΔS , E_a and rate are **often a separate mark**. Write **$\text{J K}^{-1} \text{mol}^{-1}$** , not " J/K mol ". **pH has no units.**

Plausibility checks: weak-acid pH **between 1 and 7**; excess-base pH **> 7**; buffer pH **near pKa**.

Wrong chemistry alongside right chemistry is penalised: "atoms" for ions · "chlorine" for chloride · "**absorb**" for **adsorb** · "clear" for colourless · "intermolecular forces" for **ion-dipole**.

Names: exact IUPAC — "**propan-1-ol**", "**2-hydroxypropanenitrile**". If you give both a name and a formula, **both must be right**.

Graphs: axes the right way round, labelled with units (**ln k has none; $1/T$ in K^{-1}**), $\geq$ half the grid, points $\pm \frac{1}{2}$ square, **best-fit line not point-to-point, working shown on the graph**.